「Thm.

For every continuous map $f: S^1 \to \mathbb{R}^1$, there exists $x \in S^1$ such that $f(x) = f(-x)$.

Proof. Let $h(x) := f(x) - f(-x)$. If h is identically zero, then there is nothing to prove.

Assume that $h(x_0) \neq 0$ for some $x_0 \in S^1$.

If $h(x_0) > 0$, then $h(-x_0) = f(-x_0) - f(x_0) = -h(x_0) < 0$,

by the intermidate property for a continuous map h, for some t,

$$h(t) = f(t) - f(-t) = 0$$

Similarly, if $h(x_0) < 0$, we can obtain the same result. ☒

「Borsuk-Ulam Theorem.

「Thm.

For every continuous map $f: S^2 \to \mathbb{R}^2$, there exists $x \in S^2$ such that $f(x) = f(-x)$.

Proof. Suppose that there exists a continuous map $f: S^2 \to \mathbb{R}^2$ satisfying for all $x \in S^2$, $f(x) \neq f(-x)$. Then, a function

$$g: S^2 \to S^1;\ x \mapsto \frac{f(x) - f(-x)}{|f(x) - f(-x)|}$$

is well-defined. Define a loop

$$\eta: I \to S^2 (\subset \mathbb{R}^3);\ t \mapsto (\cos 2\pi t, \sin 2\pi t, 0).$$